CBSE Class 10 Maths Basic Question Paper 2024

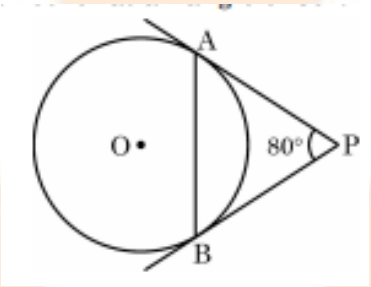
Set – 1 (430/3/1) Solutions

Section - A

(Multiple Choice Questions)

Q. No. 1 to 20 are Multiple Choice Questions of 1 mark each. 20 x 1=20

1. In the given figure, tangents PA and PB drawn from P to circle are inclined to each other at an angle of 80°. The measure of $\angle$PAB is 1



(A) 80° **(B) 60°**

(C) 50° **(D) 40°**

Ans: (C) 50°

2. The value(s) of k for which the quadratic equation $5x^2 - 9kx + 5 = 0$ has real and equal roots, is/are: 1

(A) $\frac{-10}{9}$ **(B)** $\pm\frac{9}{10}$

(C) $\frac{10}{9}$

(D) $\pm\frac{10}{9}$

Ans: (B) $\pm\frac{9}{10}$

3. The distance between the points A(-1, 5) and B(6, -2) is: **1**

(A) $2\sqrt{7}$

(B) $7\sqrt{2}$

(C) 49

(D) 14

Ans: (B) $7\sqrt{2}$

4. A quadratic polynomial whose zeroes are 3 and -2, is: **1**

(A) $x^2 - x - 6$

(B) $x^2 + x - 6$

(C) $2x^2 - x - 12$

(D) $x^2 + x + 6$

Ans: (A) $x^2 - x - 6$

5. The lines represented by linear equations xa and y = b (a $\neq$ b) are **1**

(A) intersecting at (a, b).

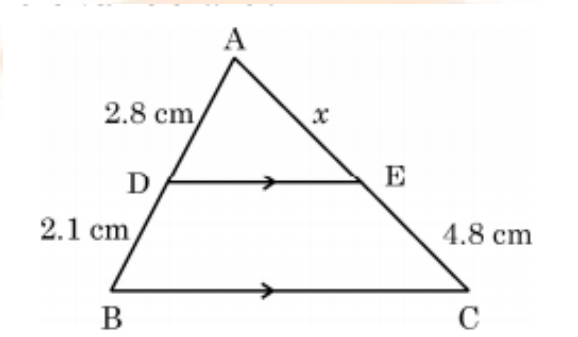
(B) intersecting at (b, a).

(C) parallel.

(D) coincident.

Ans: (A) intersecting at (a, b)

6. If in the given figure, DE || BC. If AD = 2.8 cm, DB=2.1 cm and EC = 4.8 cm, then the value of x is: **1**



(A) 3.6 cm **(B) 2.4 cm**

(C) 6.4 cm **(D) 4.8 cm**

Ans: (C) 6.4 cm

7. In a right-angled triangle ABC, $\angle A = 90°$ and AB = AC. The value of sin C is: **1**

(A) 0 **(B)** $\frac{\sqrt{3}}{2}$

(C) $\frac{1}{2}$ **(D)** $\frac{1}{\sqrt{2}}$

Ans: (D) $\frac{1}{\sqrt{2}}$

8. A die is thrown once. The probability of getting a number less than 6, is: **1**

(A) 0 **(B)** $\frac{5}{6}$

(C) $\frac{1}{6}$ **(D) 1**

Ans: (B) $\frac{5}{6}$

9. From an external point P, a tangent PA is drawn to a circle. The number of tangents through P parallel to PA is: 1

(A) 2 **(B) more than 2**

(C) 1 **(D) 0**

Ans: (D) 0

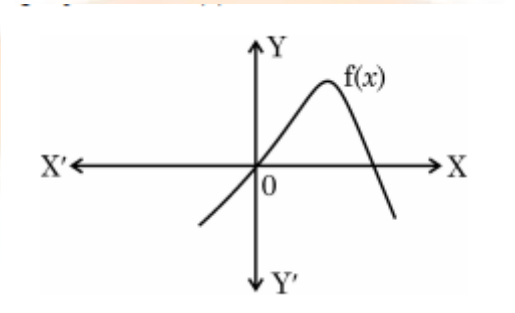
10. If the volume of a sphere is $\frac{11}{21} cm^3$, then the radius of the sphere is: 1

(A) 2 cm **(B) 4 cm**

(C) $\frac{1}{2}$ **cm** **(D)** $\frac{1}{4}$ **cm**

Ans: (C) $\frac{1}{2}$ cm

11. In the given figure, graph of a polynomial f(x) is shown. The number of zeroes of polynomial f(x) is: 1

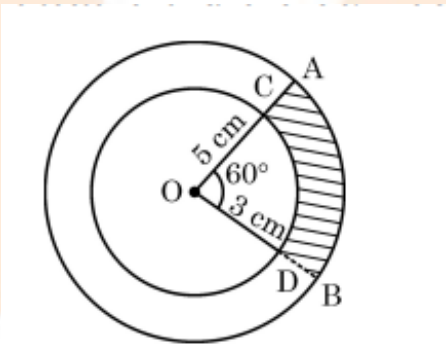


(A) 3 **(B) 1**

(C) 0 **(D) 2**

Ans: (D) 2

12. In the given figure, two concentric circles of radii 5 cm and 3 cm have their centre O. OAB is a sector of outer circle making an angle of 60° at the centre while OCD is the sector of smaller circle. The area of the shaded region is: 1



(A) $\frac{7\pi}{2} cm^2$ **(B)** $\frac{8\pi}{3} cm^2$

(C) $\frac{25\pi}{6} cm^2$

(D) $\frac{3\pi}{2} cm^2$

Ans: (B) $\frac{8\pi}{3} cm^2$

13. If the mean and median of a data are 10 and 11 respectively, then mode of the data is: 1

(A) 12

(B) 8

(C) 20

(D) 13

Ans: (D) 13

14. Two fair coins are tossed together. The probability of getting 2 heads, is: 1

(A) $\frac{1}{2}$

(B) $\frac{3}{4}$

(C) $\frac{1}{4}$

(D) $\frac{3}{2}$

Ans: (C) $\frac{1}{4}$

15. If cos A = $\frac{1}{2}$**, then tan A is equal to 1**

(A) $\frac{1}{\sqrt{3}}$

(B) $\sqrt{3}$

(C) 3 **(D)** $\frac{3}{2}$

Ans: (B) $\sqrt{3}$

16. From a solid cube of side 14 cm, a sphere of maximum diameter is carved out. The radius of sphere is: 1

(A) 7cm **(B) 14cm**

(C) $\frac{7}{2}$ **cm** **(D)** $\sqrt{14}$ **cm**

Ans: (A) 7cm

17. The common difference of an A.P. whose nth term is given by a = 5n - 1, is: 1

(A) 1 **(B) 6**

(C) 5 **(D) 4**

Ans: (C) 5

18. If for a distribution, $\sum_{1}^{2} f_i x_i = 132 + 5p$, $\sum_{1}^{n} f_i = 20$ and the mean of the distribution is 8.1, then the value of p is: **1**

(A) 3 **(B) 6**

(C) 4 **(D) 5**

Ans: (B) 6

(Assertion – Reason based questions)

Directions : In question numbers 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option from the following :

(A) Both Assertion (A) and Reason (R) are true. Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true. Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

19. Assertion (A) : The distance of P(a, b) from origin is $a^2 + b^2$. **1**

Reason (R) : The distance between two points A(x_1, x_2) and B(y_1, y_2) is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Ans: (D) Assertion (A) is false, but Reason (R) is true.

20. Assertion (A) : $\sqrt{2}(5-\sqrt{2})$ is an irrational number. **1**

Reason (R) : Product of two irrational numbers is always irrational.

Ans: (C) Assertion (A) is true, but Reason (R) is false.

Section – B

(Very Short Answer Type Questions)

Q. No. 21 to 25 are Very Short Answer type questions of 2 marks each.

21. (a) If Q(0, 2) is equidistant from P(5, –3) and R(*x*, 7), find the value(s) of *x*. **2**

OR

(b) If A(1, 1) and B(7, 9) are the end points of a diameter of a circle, then find the co-ordinates of the centre of the circle. **2**

Ans:

(a) The distance formula between two points (x_1, y_1) and (x_2, y_2) is given by:

d = $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

For Q(0, 2) and P(5, -3):

$d_{QP} = \sqrt{(5-0)^2+(-3-2)^2}$

$d_{QP} = \sqrt{5^2+(-5)^2}$

$d_{QP} = \sqrt{25+25}$

$d_{QP} = \sqrt{50}$

$d_{QP} = 5\sqrt{2}$

For Q(0, 2) and R(x, 7):

$d_{QR} = \sqrt{(x-0)^2 + (7-2)^2}$

$d_{QR} = \sqrt{x^2 + 5^2}$

$d_{QR} = \sqrt{x^2 + 25}$

Set the distances equal to each other since Q is equidistant from P and R:

$d_{QP} = d_{QR}$

$5\sqrt{2} = \sqrt{x^2 + 25}$

$(5\sqrt{2})^2 = (\sqrt{x^2 + 25})^2$

$50 = x^2 + 25$

$x^2 = 50 - 25$

$x^2 = 25$

$x = \pm 5$

The values of x are 5 and -5.

OR

(b) The midpoint $M(x_m, y_m)$ of a line segment with endpoints (x_1, y_1) and (x_2, y_2) is given by:

$$M(x_m, y_m) = \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}$$

Apply the same formula to the given points:

For the endpoints A(1, 1) and B(7, 9):

$$x_m = \frac{1+7}{2} = \frac{8}{2} = 4$$

$$y_m = \frac{1+9}{2} = \frac{10}{2} = 5$$

The coordinates of the center of the circle are (4, 5).

22. Evaluate: $4\sin^2 60^\circ \tan^2 45^\circ - 2\sec^2 30^\circ \tan^2 60^\circ$ **2**

Ans: Given $4\sin^2 60^\circ \tan^2 45^\circ - 2\sec^2 30^\circ \tan^2 60^\circ$

We know that $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\tan 45^\circ = 1$, $\sec 30^\circ = \frac{2}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$

Substitute those values in given equation

$$\Rightarrow 4\left(\frac{\sqrt{3}}{2}\right)^2 \times (1)^2 - 2\left(\frac{2}{\sqrt{3}}\right)^2 \times (\sqrt{3})^2$$

$$\Rightarrow 4\left(\frac{3}{4}\right) \times 1 - 2\left(\frac{4}{3}\right) \times 3$$

$$\Rightarrow 3 - 8 = -5$$

23. (a) Prove that $-7-2\sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is an irrational number. **2**

OR

(b) Explain why (7 × 11 × 13 + 2 × 11) is not a prime number. **2**

Ans:

(a)

1. Assume the Opposite:

Suppose $-7-2\sqrt{3}$ is a rational number. This means it can be written as a fraction $\frac{p}{q}$, where p and q are integers and $q \neq 0$.

2. Isolate $\sqrt{3}$:

If $-7-2\sqrt{3} = \frac{p}{q}$, then:

$$-2\sqrt{3} = \frac{p}{q} + 7$$

$$\sqrt{3} = -\frac{1}{2}\frac{p}{q} + 7$$

3. Rational Result: The right side of the equation, $-\frac{1}{2}(\frac{p}{q}+7)$, is a rational number because it's a combination of rational numbers. So, $\sqrt{3}$ would be rational if our assumption were true.

4. Contradiction:

But we know that $\sqrt{3}$ is actually irrational. This contradiction shows that our assumption must be false.

Therefore, $-7-2\sqrt{3}$ cannot be rational and must be irrational.

OR

(b) $7\times11\times13+2\times11$

$= 11\times(7\times13+2)$

$= 11\times(91+2)$

$= 11\times93$

$= 1023$

A prime number can only be divided by 1 and itself. Here 1023, we can divide it by 11 and 93, besides 1 and the number itself.

Since 1023 has factors other than 1 and itself, it is not a prime number.

24. Find the ratio in which the Y-axis divides the line segment joining the points A(5, –6) and B(–1, –4). Also, find the point of intersection. 2

Ans: The Y-axis has x = 0. To find where the line intersects the Y-axis, use the formula for a point dividing a line segment:

$$\frac{kx_2+x_1}{k+1}, \frac{ky_2+y_1}{k+1}$$

Put x = 0:

$$0 = \frac{k\times(-1)+5}{k+1}$$

$0 = -k + 5 \Rightarrow k = 5$

So, the Y-axis divides the line segment in a ratio of 5:1.

Substitute k = 5 into the section formula for y:

$$y = \frac{5\times(-4)+(-6)}{5+1}$$

$$y = \frac{-20-6}{6} = \frac{-26}{6} = -\frac{13}{3}$$

Therefore, the point of intersection is $(0, -\frac{13}{3})$.

25. There are 80 cards numbered from 1 to 80. One card is drawn at random from them. Find the probability that the number on the selected card is not divisible by 8. 2

Ans: There are 80 cards, so the total number of possible outcomes is 80.

$$\frac{80}{8} = 10$$

So, there are 10 cards with numbers divisible by 8 (8, 16, 24, 32, 40, 48, 56, 64, 72, 80).

Subtract the number of cards divisible by 8 from the total number of cards:

80 - 10 = 70

So, there are 70 cards that are not divisible by 8.

The probability is the number of favorable outcomes (cards not divisible by 8) divided by the total number of outcomes (total cards):

$$\text{Probability} = \frac{\text{Number of cards not divisible by 8}}{\text{Total number of cards}} = \frac{70}{80}$$

$$\frac{70}{80} = \frac{7}{8}$$

The probability that the number on the selected card is not divisible by 8 is $\frac{7}{8}$.

Section – C

(Short Answer Type Questions)

Q. No. 26 to 31 are Short Answer type questions of 3 marks each.

26. Find LCM and HCF of two numbers 336 and 54, using prime-factorisation method.

3

Ans:

For 336:

$$\begin{array}{r|l} 2 & 336 \\ 2 & 168 \\ 2 & 84 \\ 2 & 42 \\ 2 & 21 \\ 3 & 7 \end{array}$$

Prime factorisation of 336:

$336 = 2^4 \times 3^1 \times 7^1$

For 54:

$2 \underline{|54}$
$2 \underline{|27}$
$3 \underline{|9}$
$3 \underline{|3}$

Prime factorisation of 54:

$54 = 2^1 \times 3^3$

The HCF is found by taking the lowest power of all common prime factors:

- Common prime factors: 2 and 3

- Lowest power of 2: 2^1

- Lowest power of 3: 3^1

So, the HCF is:

HCF = $2^1 \times 3^1 = 2 \times 3 = 6$

The LCM is found by taking the highest power of all prime factors present in either number:

- Highest power of 2: 2^4

- Highest power of 3: 3^3

- Highest power of 7: 7^1 (only in 336)

So, the LCM is:

LCM = $2^4 \times 3^3 \times 7^1$

= 16 × 27 × 7

= 16 × 189

= 3024

HCF of 336 and 54 is 6.

LCM of 336 and 54 is 3024.

27. The altitude of a right-angled triangle is 7 cm less than its base. If its hypotenuse is 17 cm long, then **3**

(a) represent the above information in the form of a quadratic equation;

(b) find the length of the sides of the triangle.

Ans:

(a) Let the base of the right-angled triangle be x cm.

The altitude is 7 cm less than the base, so it is x − 7 cm.

The hypotenuse is 17 cm.

Using the Pythagorean theorem, which states that in a right-angled triangle:

$(\text{Base})^2 + (\text{Altitude})^2 = (\text{Hypotenuse})^2$

Substitute the given values:

$$x^2 + (x-7)^2 = (17)^2$$

$$x^2 + (x-7)^2 = 289$$

$$x^2 + (x^2 + 49 - 14x) = 289$$

$2x^2 - 14x + 49 = 289$

$2x^2 - 14x + 49 - 289 = 0$

$2x^2 - 14x - 240 = 0$

Divide the entire equation by 2:

$x^2 - 7x - 120 = 0$

So, the quadratic equation representing the problem is: $x^2 - 7x - 120 = 0$

(b) The equation is $x^2 - 7x - 120 = 0$

$x^2 + 8x - 15x - 120 = 0$

$x(x+8) - 15(x+8) = 0$

(x + 8) (x – 15) = 0

Therefore, x= 15 (as x $\neq$ -8)

$\therefore$ The sides of the triangle other than hypotenuse are 15 cm and 8 cm.

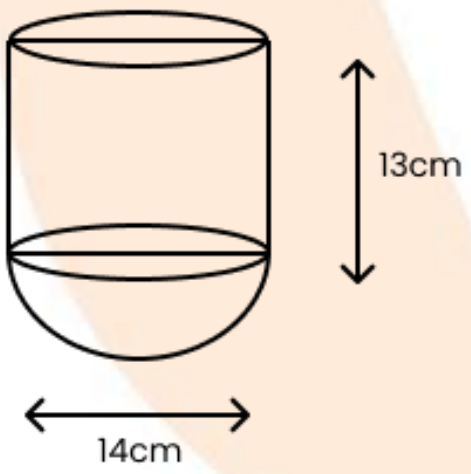
28. (a) A vessel is in the form of a hollow hemisphere surmounted by a hollow cylinder. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find the inner surface area of the vessel. 3

OR

(b) A solid toy is in the form of a hemisphere surmounted by a right circular cone. The height of the cone is 2 cm and the diameter of the base is 4 cm. Determine the volume of the toy. 3

Ans:

(a)



Given:

Diameter of the hemisphere d = 14 cm, so the radius $r = \frac{d}{2} = 7$ cm.

Total height of the vessel h_{total} = 13 cm.

The height of the cylindrical part is the total height of the vessel minus the radius of the hemisphere.

$h_{cylinder} = h_{total}$ - r = 13 - 7 = 6 cm

The inner surface area of the hemisphere (excluding the base) is given by:

Surface Area of Hemisphere $= 2\pi r^2$

Substituting r = 7 cm:

Surface Area of Hemisphere $= 2\pi \times (7)^2 = 2\pi \times 49 = 98\pi$ square cm

The inner surface area of the cylinder (excluding the top and bottom) is given by:

Surface Area of Cylinder $= 2\pi rh$

Substituting r = 7 cm and h = 6 cm:

Surface Area of Cylinder $= 2\pi \times 7 \times 6 = 84\ \pi$ square cm

The total inner surface area of the vessel is the sum of the inner surface areas of the hemisphere and the cylinder:

Total Inner Surface Area = Surface Area of Hemisphere + Surface Area of Cylinder

Total Inner Surface Area $= 98\pi + 84\pi = 182\pi$ square cm

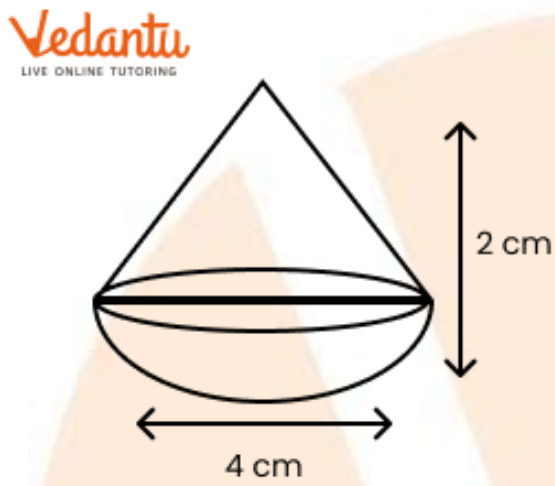
Substitute $\pi = 3.17$

Total inner surface Area = 182(3.14) = 571.4 sq. cm

The inner surface area of the vessel is 571.4 sq. cm.

OR

(b)



Given:

Height of the cone h_{cone} = 2 cm

Diameter of the base of the cone d = 4 cm, so the radius r = $\frac{d}{2}$ = 2 cm

The volume V_{cone} of a right circular cone is given by:

$$V_{cone} = \frac{1}{3}\pi r^2 h_{cone}$$

Substituting r = 2 cm and h_{cone} = 2 cm:

$$V_{cone} = \frac{1}{3}\pi(2)^2 \times 2 = \frac{1}{3}\pi \times 4 \times 2 = \frac{8\pi}{3} \text{ cubic cm}$$

The radius of the hemisphere is the same as the radius of the cone's base, which is r = 2 cm.

The volume $V_{hemisphere}$ of a hemisphere is given by:

$V_{hemisphere} = \frac{2}{3}\pi r^3$

Substituting r = 2 cm:

$V_{hemisphere} = \frac{2}{3}\pi(2)^3 = \frac{2}{3}\pi \times 8 = \frac{16\pi}{3}$ cubic cm

The total volume V_{toy} is the sum of the volumes of the hemisphere and the cone:

$V_{toy} = V_{hemisphere} + V_{cone}$

$V_{toy} = \frac{16\pi}{3} + \frac{8\pi}{3} = \frac{24\pi}{3} = 8\pi$ cubic cm

Substitute value of π as 3.14

$V_{toy} = 8(3.14) = 25.12\ cm^3$

The volume of the toy is 25.12 cm^3

29. Find the zeroes of the quadratic polynomial $5x^2 + 3x - 2$ and verify the relationship between the zeroes and the co-efficients. 3

Ans: The quadratic polynomial is of the form $ax^2 + bx + c$, where:

a = 5

b = 3

c = -2

Formula to find the zeroes

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Substitute the values of a, b, and c:

$$x = \frac{-3 \pm \sqrt{3^2 - 4 \cdot 5 \cdot (-2)}}{2 \cdot 5}$$

$$x = \frac{-3 \pm \sqrt{9 + 40}}{10}$$

$$x = \frac{-3 \pm \sqrt{49}}{10}$$

$$x = \frac{-3 \pm 7}{10}$$

So, the two zeroes are:

$$x_1 = \frac{-3+7}{10} = \frac{4}{10} = \frac{2}{5}$$

$$x_2 = \frac{-3-7}{10} = \frac{-10}{10} = -1$$

For a quadratic polynomial $ax^2 + bx + c$, if α and β are the zeroes, the following relationships hold:

Sum of Zeroes: $\alpha + \beta = -\frac{b}{a}$

Product of Zeroes: $\alpha \cdot \beta = \frac{c}{a}$

Sum of Zeroes: $\alpha + \beta = \frac{2}{5} + (-1) = \frac{2}{5} - 1 = \frac{2-5}{5} = -\frac{3}{5}$

Product of Zeroes:

$\alpha \cdot \beta = \frac{2}{5} \times (-1) = -\frac{2}{5}$

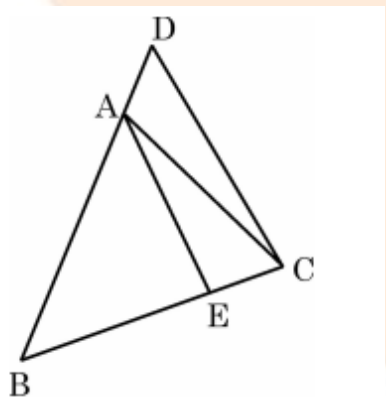
$\frac{c}{a} = \frac{-2}{5}$

The zeroes of the polynomial $5x^2 + 3x - 2$ are $\frac{2}{5}$ and -1.

The sum of the zeroes is $-\frac{3}{5}$, which matches $-\frac{b}{a}$.

The product of the zeroes is $-\frac{2}{5}$, which matches $\frac{c}{a}$.

30. In the given figure, $\angle ABC = \angle ACB$ and $\frac{BC}{BE} = \frac{BD}{AC}$ **3**



Show that ΔABE ~ ΔDBC and AE || DC.

Ans: $\Delta ABE \sim \Delta DBC$:

Given that $\angle ABC = \angle ACB$.

We know that BC / BE=BD / AC

Since $\angle ABC = \angle ACB$, it follows that $\angle ABE = \angle DBC$ because they are corresponding angles.

By the given proportion, we have:

$$\frac{BC}{BE} = \frac{BD}{AC}$$

$$\frac{AB}{BD} = \frac{BE}{BC}$$ (inverse of the given ratio)

Therefore, we have two pairs of corresponding sides in proportion and the included angle is equal:

$\angle ABE = \angle DBC$

By the Side-Angle-Side (SAS) similarity criterion, $\Delta ABE \sim \Delta DBC$.

Now, we know that $\Delta ABE \sim \Delta DBC$

This implies that corresponding angles are equal. Therefore, $\angle AEB = \angle DCB$.

Since $\angle AEB = \angle DCB$ and they are alternate interior angles formed by the transversal \(BE\) intersecting the lines AE and DC, it follows that:

AE || DC

31. (a) Prove that $(\sin\theta+\operatorname{cosec}\theta)^2+(\cos\theta+\sec\theta)^2=7+\tan^2\theta+\cot^2\theta$**.** **3**

OR

(b) If cos A = $\frac{5}{13}$**, then verify that** $\frac{\cos A}{1-\tan A}+\frac{\sin A}{1-\cot A}$ **= cos A + sin A.** **3**

Ans:

(a) $(\sin\theta+\operatorname{cosec}\theta)^2+(\cos\theta+\sec\theta)^2 = 7+\tan^2\theta+\cot^2\theta$

L.H.S= $(\sin\theta+\operatorname{cosec}\theta)^2+(\cos\theta+\sec\theta)^2$

First, consider $(\sin\theta+\operatorname{cosec}\theta)^2$:

$(\sin\theta+\operatorname{cosec}\theta)^2 = \sin^2\theta+2\sin\theta\operatorname{cosec}\theta+\operatorname{cosec}^2\theta$

Since $\operatorname{cosec}\theta=\frac{1}{\sin\theta}$, we have:

$\sin^2\theta+2\sin\theta\frac{1}{\sin\theta}+\frac{1}{\sin^2\theta}$

$\sin^2\theta+2+\frac{1}{\sin^2\theta}$

$\sin^2\theta+2+\operatorname{cosec}^2\theta$

$\therefore\ (\sin\theta+\operatorname{cosec}\theta)^2=\sin^2\theta+2+\operatorname{cosec}^2\theta$

Next, consider $(\cos\theta+\sec\theta)^2$:

$(\cos\theta+\sec\theta)^2=\cos^2\theta+2\cos\theta\sec\theta+\sec^2\theta$

Since $\sec\theta = \frac{1}{\cos\theta}$, we have:

$$\cos^2\theta + 2\cos\theta\frac{1}{\cos\theta} + \frac{1}{\cos^2\theta}$$

$$\cos^2\theta + 2 + \sec^2\theta$$

So $(\cos\theta + \sec\theta)^2 = \cos^2\theta + 2 + \sec^2\theta$

Now, combine both expressions:

$$(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 = (\sin^2\theta + 2 + \operatorname{cosec}^2\theta) + (\cos^2\theta + 2 + \sec^2\theta)$$

$$= \sin^2\theta + \cos^2\theta + 2 + \operatorname{cosec}^2\theta + 2 + \sec^2\theta$$

$$= (\sin^2\theta + \cos^2\theta) + 4 + \operatorname{cosec}^2\theta + \sec^2\theta$$

Since $\sin^2\theta + \cos^2\theta = 1$:

$$= 1 + 4 + \operatorname{cosec}^2\theta + \sec^2\theta$$

$$= 5 + \operatorname{cosec}^2\theta + \sec^2\theta$$

We know that $\operatorname{cosec}^2\theta = 1 + \cot^2\theta$

$$\sec^2\theta = 1 + \tan^2\theta$$

Substitute these identities in the above equation:

$$5 + \csc^2\theta + \sec^2\theta = 5 + (1 + \cot^2\theta) + (1 + \tan^2\theta)$$

$$= 5 + 1 + \cot^2\theta + 1 + \tan^2\theta$$

$$= 7 + \tan^2\theta + \cot^2\theta$$

Therefore, $(\sin\theta + \csc\theta)^2 + (\cos\theta + \sec\theta)^2 = 7 + \tan^2\theta + \cot^2\theta$

OR

(b) $\frac{\cos A}{1-\tan A}+\frac{\sin A}{1-\cot A}=\cos A+\sin A$

Given that $\cos A=\frac{5}{13}$,

We use the Pythagorean identity:

$\sin^2 A+\cos^2 A=1$

So,

$$\sin^2 A = 1-\cos^2 A = 1-\left(\frac{5}{13}\right)^2 = 1-\frac{25}{169} = \frac{169-25}{169} = \frac{144}{169}$$

Therefore,

$$\sin A = \sqrt{\frac{144}{169}} = \frac{12}{13}$$

$$\tan A = \frac{\sin A}{\cos A} = \frac{\frac{12}{13}}{\frac{5}{13}} = \frac{12}{5}$$

$$\cot A = \frac{\cos A}{\sin A} = \frac{\frac{5}{13}}{\frac{12}{13}} = \frac{5}{12}$$

Substitute $\cos A$, $\sin A$, $\tan A$, and $\cot A into the L.H.S:

$$\frac{\cos A}{1-\tan A}+\frac{\sin A}{1-\cot A}$$

Substitute the known values:

$$\frac{\frac{5}{13}}{1-\frac{12}{5}}+\frac{\frac{12}{13}}{1-\frac{5}{12}}$$

Simplify each term separately:

For the first term:

$$\frac{\frac{5}{13}}{1-\frac{12}{5}}=\frac{\frac{5}{13}}{\frac{5-12}{5}}=\frac{\frac{5}{13}}{\frac{-7}{5}}=\frac{5}{13}\times\frac{5}{-7}=\frac{25}{-91}=-\frac{25}{91}$$

For the second term:

$$\frac{\frac{12}{13}}{1-\frac{5}{12}}=\frac{\frac{12}{13}}{\frac{12-5}{12}}=\frac{\frac{12}{13}}{\frac{7}{12}}=\frac{12}{13}\times\frac{12}{7}=\frac{144}{91}$$

Combine the two terms:

$$-\frac{25}{91}+\frac{144}{91}=\frac{-25+144}{91}=\frac{119}{91}=\frac{17}{13}$$

Since $\cos A+\sin A=\frac{5}{13}+\frac{12}{13}=\frac{17}{13}$

$$\frac{\cos A}{1-\tan A}+\frac{\sin A}{1-\cot A}=\cos A+\sin A$$

Section – D

(Long Answer Type Questions)

Q. No. 32 to 35 are Long Answer type questions of 5 marks each.

32. Consider the following distribution of hourly wages of 50 workers of a factory: 5

Hourly wages (in ₹)	**100 – 120**	**120 – 140**	**140 – 160**	**160 – 180**	**180 – 200**
Number of workers	**12**	**14**	**8**	**6**	**10**

Find the mean and the median of the above data.

Ans:

C.I	**X**	**f**	**u =** $\frac{x-150}{20}$	**fu**	**cf**
100 – 120	110	12	-2	-24	12
120 – 140	130	14	-1	-14	26
140 – 160	150	8	0	0	34
160 – 180	170	6	1	6	40
180 – 200	190	10	2	20	50
		50		-12	

Mean = 150 - $\frac{12}{50}\times 20$

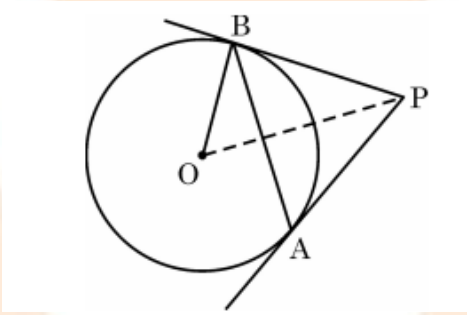
Mean = 150 – 4.8 = 145.2

Median = $120 + \frac{20}{14}(25-12)$

Median = 120 + 1.42(13)

Median = 120 +18.46 = 138.46

33. (a) In the given figure, AB is chord of length 6 cm of a circle of radius 5 cm. The tangents at A and B intersect at a point P. Find the length of PB. 5

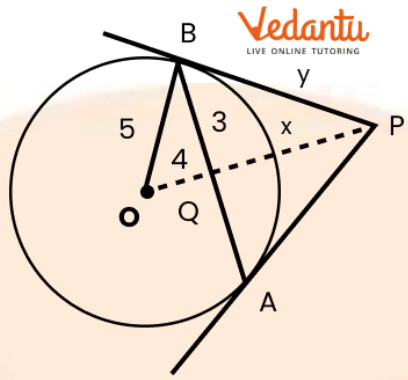


OR

(b) Prove that the parallelogram circumscribing a circle is a rhombus. Also, find area of the rhombus, if radius of circle is 3 cm and length of one side of the rhombus is 10 cm. 5

Ans:

(a)



From the given figure OP $\perp$ AB and bisects it

Given AB = 6 cm, radius (OB) = 5 cm

$\therefore$ BQ = QA = 3 cm

ΔOQB is a right angled- triangle so $OB^2 = OQ^2 + BQ^2$

Substitute the known values

$5^2 = OQ^2 + 3^2$

$25 = OQ^2 + 9$

$OQ^2 = 16$

OQ = 4 cm

Using Pythagoras theorem in ΔOBP and ΔPQB

$y^2 = 3^2 + x^2 \ldots..(1)$ and $(x+4)^2 = y^2 + 25 \ldots.(2)$

Substitute y^2 in (2)

$(x+4)^2 = 3^2 + x^2 + 25$

$x^2 + 16 + 8x = 9 + x^2 + 25$

$8x = 34 - 16$

8x = 18

$x = \frac{18}{8} = \frac{9}{4}$

Substitute x to get y in (1)

$y^2 = 9 + \left(\frac{9}{4}\right)^2$

$y^2 = 9 + \left(\frac{81}{16}\right)$

$y^2 = \frac{144 + 81}{16}$

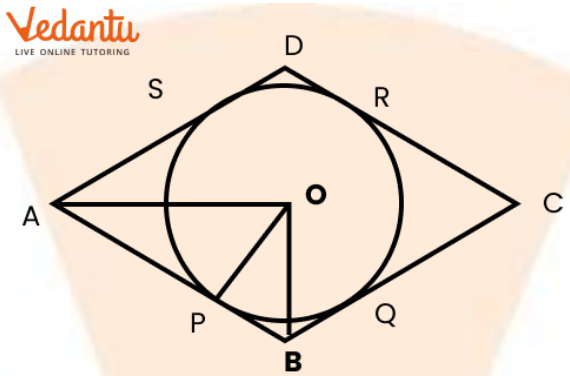
$y^2 = \frac{225}{16}$

$y = \sqrt{\frac{225}{16}} = \frac{15}{4}$

Therefore, the length of PB(y) $= \frac{15}{4}$

OR

(b)



Let ABCD be a parallelogram touching the circle at P, Q, R and S

So that AP = AS, BP = BQ, CR = CQ and DR = DS

Now add all the equations

AP + BP + CR + DR = AS + BQ + CQ + DS

AB + CD = AD + BC

2 AB = 2 BC (from figure)

AB = BC

Hence, ABCD us a rhombus

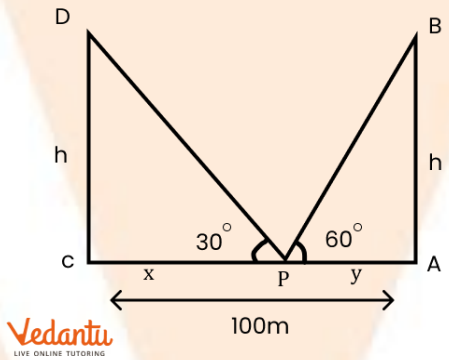
Area of rhombus = 4 $\times$ area of triangle AOB

$$= 4 \times \frac{1}{2} \times \text{base} \times \text{height}$$

$$= 2 \times 10 \times 3 = 60 \ cm^2$$

34. Two poles of equal height are standing opposite each other on either side of a road, which is 100 m wide. From a point somewhere between them on the road, the angles of elevation of the top of the poles are 60º and 30º respectively. Find the height of the poles and the distances of the point from the poles. 5

Ans:



The height of each pole as h.

The distance from the point to the pole with the 60º angle of elevation as x.

The distance from the point to the pole with the 30º angle of elevation as 100 - x.

From the point on the road, the angles of elevation to the tops of the poles are 60º and 30º. Using the tangent of these angles, we get:

For the pole with the 60º angle of elevation:

$$\tan 60^\circ = \frac{h}{x}$$

Since $\tan 60^\circ = \sqrt{3}$:

$$\sqrt{3} = \frac{h}{x}$$

$h = \sqrt{3}x \quad \ldots(1)$

For the pole with the 30º angle of elevation:

$$\tan 30^\circ = \frac{h}{100 - x}$$

Since $\tan 30^\circ = \frac{1}{\sqrt{3}}$:

$$\frac{1}{\sqrt{3}} = \frac{h}{100 - x}$$

$$h = \frac{100 - x}{\sqrt{3}} \quad \ldots.(2)$$

Equate the two expressions for h from Equations 1 and 2:

$$\sqrt{3}x = \frac{100 - x}{\sqrt{3}}$$

Multiply both sides by $\sqrt{3}$ to eliminate the denominator:

3x = 100 - x

Solve for x:

3x + x = 100

4x = 100

x = 25 m

Find the distance from the point to the other pole:

100 - x = 100 - 25 = 75 m

Substitute x = 25 into Equation 1:

$h = \sqrt{3} \times 25 = 25\sqrt{3}$ m

The height of each pole is $25\sqrt{3}$ meters (approximately 43.3 meters).

The distances from the point to the poles are 25 meters and 75 meters.

35. (a) Using graphical method, solve the following pair of equations: 5

***x* + 2y = 8 and 3*x* – 2y = 12**

OR

(b) The sum of the digits of a 2-digit number is 9. Also, nine times this number is twice the number obtained by reversing the order of the digits. Find the number. 5

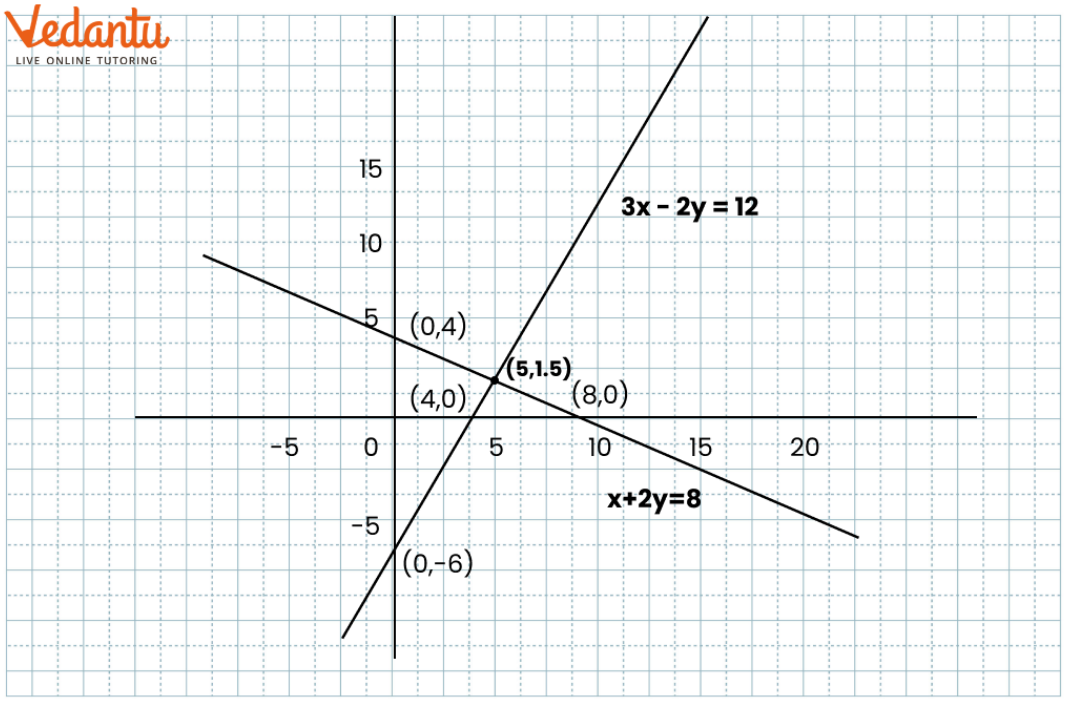
Ans:

(a) x + 2y = 8

x	0	8
y	4	0

3x – 2y = 12

x	0	4
y	-6	0



From graph we have x = 5, y = 1.5

OR

(b) Let the digit at ones place be x and tens plpace is y

From given we can say that

$x + y = 9$….(1)

$9(10y + x) = 2(10x + y)$

$90y + 9x = 20x + 2y$

$11x = 88y$

$x = 8y$

Substitute x in (1)

8y + y = 9

y = 1 and x = 8

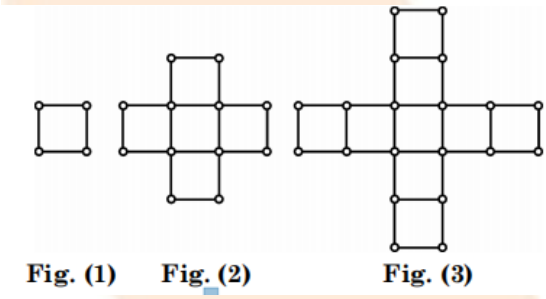
Therefore, the required number = 18

Section – E

(Case Study based Questions)

Q. No. 36 to 38 are Case Study based questions of 4 marks each.

36. While preparing for a competitive examination, Akbar came across a match-stick pattern based question. The pattern is given below:



Based on the above information, answer the following questions:

(i) Write first term and common difference of the A.P. formed by number of squares in each figure. 1

(ii) Write first term and common difference of the A.P. formed by number of sticks used in each figure. 1

(iii) (a) How many squares are there in Fig. (10) ? Also, write the number of sticks used in Fig. (10). 2

OR

(iii) (b) If 88 sticks are used to make m^{th} figure (Fig. (m)), find the value of m. How many squares are formed in this figure? 2

Ans:

(i) First Term = 1 and Common difference = 4

(ii) First Term = 4 and Common difference = 12

(iii) (a) Required number of squares = $1 + (9) \times 4 = 37$

Required number of sticks = $4 + 9 \times 12 = 112$

OR

(iii) (b) $88 = 4 + (m - 1) \times 12$

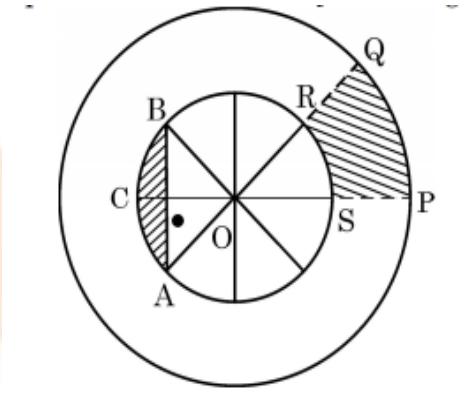
$\Rightarrow m = 8$

Number of squares formed in 8th fig. = $1 + 7 \times 4 = 29$

37. NSS (National Service Scheme) aims to connect the students to the community and to involve them in problem solving process.

NSS symbol is based on the 'Rath' wheel of the Konark Sun Temple situated in Odisha. The wheel signifies the progress cycle of life.

The diagramatic representation of the symbol is given below:



Observe the figure given above. The diameters of inner circle are equally placed. Given that OP = 21 cm, OS = 10 cm.

Based on the above information, answer the following questions :

(i) Find m∠ROS. **1**

(ii) Find the perimeter of sector OPQ. **1**

(iii) (a) Find the area of shaded region PQRS. **2**

OR

(iii) (b) Find the area of shaded region ACB i.e. the segment ACB. **2**

Ans:

(i) $\angle ROS = \frac{360^\circ}{8} = 45^\circ$

(ii) Perimeter of sector OPQ $= 21 + 21 + \frac{45}{360} \times 2 \times \frac{22}{7} \times 21 = 58.5$ cm

(iii) (a) Area of the region PQRS = Ar (OPQO) – Ar (OSRO)

$$= \frac{45}{360} \times \frac{22}{7} \times (21^2 - 10^2)$$

$$= \frac{1}{8} \times \frac{22}{7} \times 341$$

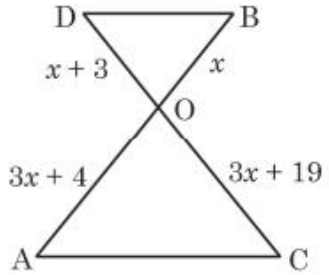
$$= \frac{3751}{28} \text{ cm2 or } 133.96 \text{ cm2}$$

OR

(iii) (b) Area of segment ACB = $\frac{90}{360} \times \frac{22}{7} \times 100 - \frac{1}{2} \times 10 \times 10$

$$= \frac{200}{7}\ cm^2 \textbf{ or } \mathbf{28.57}\ cm^2$$

38. In the figure given below, a folding table is shown:



The legs of the table are represented by line segments AB and CD intersecting at O. Join AC and BD.

Considering table top is parallel to the ground, and OB = x, OD = x + 3, OC = 3x + 19 and OA = 3x + 4, answer the following questions:

(i) Prove that ΔOAC is similar to ΔOBD. **1**

(ii) Prove that $\frac{OA}{AC} = \frac{OB}{BD}$. **1**

(iii) (a) Observe the figure and find the value of *x*. Hence, find the length of OC. **2**

OR

(iii) (b) Observe the figure and find $\frac{BD}{AC}$. **2**

Ans:

(i) Since ∠D = ∠C and ∠B = ∠A (Alternate interior angles)

∴ Δ OAC ~ Δ OBD (By AA similarity)

(ii) Δ OAC ~ Δ OBD ⇒ $\frac{OA}{OB} = \frac{AC}{BD}$ **or** $\frac{OA}{AC} = \frac{OB}{BD}$

(iii) (a) Δ OAC ~ Δ OBD ⇒ $\frac{OA}{OB} = \frac{OC}{OD}$

$$\Rightarrow \frac{3x+4}{x} = \frac{3x+19}{x+3}$$

$$\Rightarrow (x+3)(3x+4) = (3x+19)(x)$$

$$\Rightarrow 3x^2 + 4x + 9x + 12 = 3x^2 + 19x$$
$$\Rightarrow 13x + 12 - 19x = 0$$
$$\Rightarrow 12 - 6x = 0$$
$$x = 2$$

Therefore OC = 3(2) + 19 = 6 + 19 = 25

OR

(b) Δ OBD ~ Δ OAC $\Rightarrow \frac{OB}{OA} = \frac{OD}{OC} = \frac{BD}{AC}$

$$\Rightarrow \frac{x}{3x+4} = \frac{x+3}{3x+19}$$

$$x(3x+19) = (x+3)(3x+4)$$

From above answer we get $x = 2$

$$\therefore \frac{BD}{AC} = \frac{2}{10} = \frac{1}{5}$$